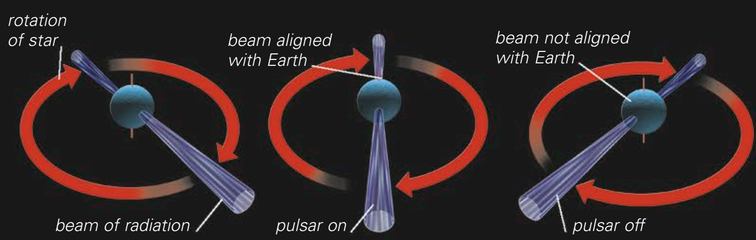
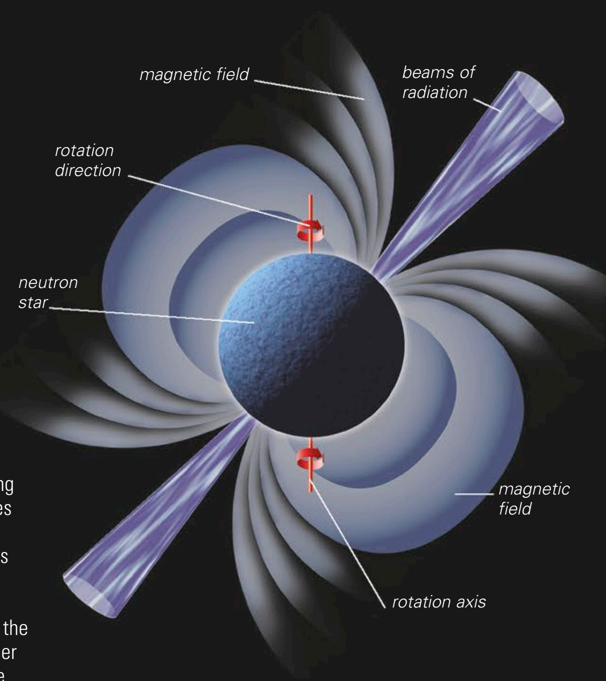
NEUTRON STARS

Neutron stars are one of the byproducts of type II supernovae explosions. During an explosion, the outer layers of a star are blown off, leaving an extremely dense, compact star consisting predominantly of neutrons with a smaller amount of electrons and protons. Neutron stars have a mass between 0.1 and 3 solar masses. Beyond this limit, a star will collapse further to become a black hole (below). As the neutron star forms, the magnetic field of the parent star becomes concentrated and grows in strength. Similarly, the original rotation of the star increases in speed as the star collapses. Neutron stars are characterized by their strong magnetic fields and rapid rotation. Over time, their rotation slows as they lose energy. However, some neutron stars show a temporary rise in rotation rate, possibly due to tremors (known as starquakes), in their thin, crystalline outer crusts. Neutron stars that emit directed pulses of radiation at regular intervals are known as pulsars (below).



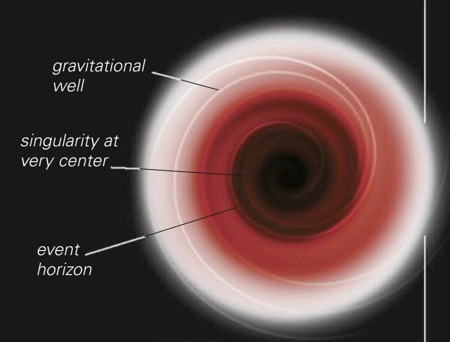
HOW PULSARS WORK
Charged particles spiral along the star's magnetic field lines and produce a beam of radiation. If the beam passes across the field of Earth, it can be detected as a pulse. Depending on the energy of the radiation, this can be in either the radio or X-ray part of the electromagnetic spectrum.



BLACK HOLES

If the remnant of a supernova explosion is greater than about 3 solar masses, there is no mechanism that can stop it from collapsing. It becomes so small and dense that its resulting gravitational pull is great enough to stop even radiation, including visible light, from escaping. Stellar-mass black holes, as such objects are known, can be detected only by the effect they have on objects around them. Light from far-off objects can be bent around a black hole as it acts as a gravitational lens, while the movement of nearby objects can be affected by a black hole's strong gravitational field (see pp.42–43). If a stellar-mass black hole is a member of a close binary system (see pp.274–275), the material from its companion star will be pulled toward it by its immense gravity. Matter will not fall directly onto the black hole due to its rotational motion. Instead, it will first be pulled into a accretion disk around the black hole. Matter impacts onto this disk, creating hot spots that can be detected by the radiation they emit. As matter in the disk gradually spirals into the black hole, friction will heat up the gas and radiation is emitted, predominantly in the X-ray part of the electromagnetic spectrum.

BLACK HOLE
Here, the gas from a companion star is drawn into a black hole via an accretion disk. When the gas crosses a limit called the event horizon, the gravitational field has become so strong that light cannot escape, and it disappears from view.



NEUTRON STAR
The gas drawn from a companion star approaches a neutron star in the same manner. However, when the gas strikes the solid surface of the neutron star, light is emitted and the star glows.